

REVISIONS			
SYM		DATE	APPROVED

PLUMBING SUMMARY		
SYSTEM & MATERIAL	FIXTURE UNITS	MAIN SIZE
DOMESTIC WATER SYSTEM	123	2"
ABOVE SLAB: TYPE "L" ANNEALED COPPER WITH 95/5 SOLDER JOINTS.		

PLUMBING LEGEND AND ABBREVIATIONS	
	COLD WATER PIPING (CW)
	HOT WATER PIPING (HW)
	HOT WATER RETURN PIPING (HWR)
	CHECK VALVE
	BALL VALVE
	SHOCK ABSORBER (SA)
	CONNECT TO EXISTING UNION
P - #	PLUMBING FIXTURE - NO.
A.F.F.	ABOVE FINISH FLOOR
FD - X	FLOOR DRAIN - TYPE (SEE SCHEDULE)
H.B.	HOSE BIBB
P.C.	PLUMBING CONTRACTOR
RPZ	REDUCED PRESSURE ZONE
WH	WALL HYDRANT
	PLAN NOTE

EXISTING PLUMBING FIXTURE SCHEDULE			
MARK	DESCRIPTION	PIPE SERVICE AND CONN. SIZE	
		CW	HW
P-1	FLR. MTD. WATER CLOSET, BACK OUTLET	1"	-
P-2	URINAL	1"	-
P-3	LAVATORY UNDERMOUNT - PROVIDE POU TMV	1/2"	1/2"
P-4	ELECTRIC WATER COOLER WITH BOTTLE FILLER	1/2"	-
P-5	MOP SINK	1/2"	1/2"
P-6	HANDWASH SINK - PROVIDE POU TMV	1/2"	1/2"
P-7	KITCHEN SINK	3/4"	3/4"
P-8	ICE MAKER	1/2"	-
P-9	STUB OUT - MISC.	1/2"	-
HB	HOSE BIB - PROVIDE NEW	3/4"	-

PROJECT NOTES
1. FIELD VERIFY FOR EXACT LOCATION OF PLUMBING FIXTURES.
2. HOSE BIBBS SHALL BE PROTECTED WITH AN APPROVED NON-REMOVABLE TYPE BACKFLOW PROTECTION DEVICE. HOSE BIBBS SHALL BE MOUNTED AT 18" ABOVE FINISH FLOOR UNLESS OTHERWISE NOTED.
3. COORDINATE AND VERIFY SIZES, LOCATIONS, DEPTHS AND PIPING PRESSURES OF ALL BUILDING UTILITIES WITH CIVIL.
4. COORDINATE AND SCHEDULE TIMING FOR UTILITY SERVICE CONNECTION WITH THE CONTRACTING OFFICER.
5. COORDINATE INSTALLATION OF ALL EQUIPMENT AND PIPING WITH OTHER TRADES PRIOR TO INSTALLATION. ENSURE THAT ALL CONTROL DEVICES, SHUT-OFF VALVES, ETC. ARE ACCESSIBLE FOR MAINTENANCE. WHERE ACCESS PANELS ARE REQUIRED IN FINISHED SPACES, OTHER THAN THAT SHOWN, CONTRACTOR SHALL PROVIDE AND COORDINATE EXACT LOCATION OF PANELS WITH ARCHITECT PRIOR TO INSTALLATION.
6. PERFORM ALL WORK IN ACCORDANCE WITH IPC AND SPECIFICATIONS.
7. THE CONTRACTOR IS RESPONSIBLE TO PROVIDE ALL MATERIALS AND LABOR NECESSARY TO COMPLETE SCOPE OF WORK UNLESS OTHERWISE SPECIFIED.
8. INSTALLATION OF PLUMBING PIPING SHALL BE COORDINATED WITH OTHER TRADES TO AVOID CONFLICTS.
9. CONTRACTOR SHALL NOT ORDER EQUIPMENT OR BEGIN FABRICATION OF PARTS PRIOR TO SHOP DRAWING APPROVAL.
10. TEST, STERILIZE AND FLUSH PIPING SYSTEMS. CLEAN ALL EQUIPMENT AND FIXTURES AT THE COMPLETION OF THE PROJECT. KEEP PREMISES CLEAN DURING CONSTRUCTION.
11. PROVIDE ACCESS DOORS/PANELS AT LOCATION OF VALVES AND OTHER COMPONENTS WHERE LOCATED ABOVE HARD CEILINGS OR INSIDE WALLS.
12. ALL WATER LINES SHALL HAVE SHUT-OFF VALVES AT CONNECTIONS OF FIXTURES AND EQUIPMENT. PROVIDE DRAIN VALVES AT PIPING LOW POINTS.
13. PIPE PENETRATIONS THROUGH FIRE OR SMOKE PARTITIONS, WALLS, AND/OR FLOORS SHALL BE MADE FIRE AND SMOKE TIGHT. MAINTAIN FIRE RATING OF FLOOR AND WALL ASSEMBLIES IN ACCORDANCE WITH UL SYSTEMS. INSTALL PIPE PENETRATION ASSEMBLIES IN ACCORDANCE WITH UL LISTED MANUFACTURER'S RECOMMENDATIONS. PENETRATIONS SHALL BE IN ACCORDANCE WITH UL THROUGH-WALL DIRECTORY.
14. FIELD VERIFY CONDITIONS BEFORE STARTING CONSTRUCTION AND NOTIFY THE ARCHITECT/ENGINEER OF DISCREPANCIES WITH THE CONSTRUCTION DOCUMENTS AND/OR POTENTIAL PROBLEMS OBSERVED BEFORE COMMENCING WORK IN AFFECTED AREAS.
15. PROVIDE ALL NECESSARY HANGERS FOR SUPPORT OF HORIZONTAL AND VERTICAL PIPING IN ACCORDANCE WITH GOOD PRACTICE AND MANUFACTURER'S RECOMMENDATIONS. PROVIDE SLEEVES AND ESCUTCHEONS FOR ALL PIPING PASSING THROUGH WALLS AND FLOORS.
16. IN GENERAL, ALL PIPING SHALL BE RUN CONCEALED IN SUSPENDED CEILING AND PIPE SPACES PROVIDED UNLESS NOTED OR INDICATED OTHERWISE.

PLUMBING EQUIPMENT SCHEDULE			
TAG	TYPE	DESCRIPTION	LOCATION
D-WH-1	DOMESTIC WATER HEATER	INDOOR, FLOOR MOUNTED, 208V ELECTRIC, 50 GALLON STORAGE TYPE ELECTRIC WATER HEATER, DUAL 4500W ELEMENT, NON-SIMULTANEOUS.	MECHANICAL ROOM
D-WH-2	DOMESTIC WATER HEATER	INDOOR, FLOOR MOUNTED, 120V ELECTRIC, 15 GALLON STORAGE TYPE ELECTRIC WATER HEATER,1500W SINGLE ELEMENT.	STORAGE ROOM
D-RCP-1	DOMESTIC HOT WATER RECIRCULATING PUMP	1/25 HP, 115V, 2 GPM, 7' HEAD, IN-LINE, SINGLE STAGE WITH TIMER AND THERMOSTATIC CONTROLLER.	MECHANICAL ROOM
ET-1	DIAPHRAGM EXPANSION TANK	4 GALLON CAPACITY, FIELD-ADJUSTABLE AIR CHARGE, PRE-CHARGE TO MATCH CW PRESSURE. MAXIMUM WORKING PRESSURE OF 150 PSI AND MAXIMUM TEMPERATURE OF 180° F. MUST BE RATED FOR POTABLE WATER.	MECHANICAL ROOM
ET-2	DIAPHRAGM EXPANSION TANK	2 GALLON CAPACITY, FIELD-ADJUSTABLE AIR CHARGE, PRE-CHARGE TO MATCH CW PRESSURE. MAXIMUM WORKING PRESSURE OF 150 PSI AND MAXIMUM TEMPERATURE OF 180° F. MUST BE RATED FOR POTABLE WATER.	STORAGE ROOM
BFP-1 (2")	BACKFLOW PREVENTER (MAIN RPZ)	THE ASSEMBLY SHALL MEET THE REQUIREMENTS OF ASSE 1013 AND BE PROVIDED TO PREVENT BACKFLOW DUE TO BACKSIPHONAGE AND/OR BACKPRESSURE. THIS DEVICE WILL BE A REDUCED PRESSURE ZONE (RPZ) TYPE. BFP-1 SHALL BE INSTALLED TO MAINTAIN ADEQUATE ACCESS FOR TESTING AND MAINTENANCE PURPOSES.	MECHANICAL ROOM
TMV	THERMOSTATIC MIXING VALVE	LAVATORY THERMOSTATIC MIXING VALVE DESIGN PARAMETER: CONTROL TEMPERATURE MUST BE ADJUSTABLE BETWEEN 100F AND 120F WITH A LOCKING NUT TO PREVENT ACCIDENTAL ADJUSTMENT. MUST CONTROL DOWN TO 0.25GPM, AND MEET ASSE 1070.	P3, P6 - POINT OF USE
HB	HOSE BIB	3/4" MPT OUTLET. DO NOT PROVIDE FROSTPROOF HOSE BIBS.	EXTERIOR WALLS - MULTIPLE LOCATIONS

CONSTRUCTION DOCUMENTS				P001	
		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND			
		MARINE CORPS BASE CAMP LEJEUNE, NORTH CAROLINA			
DES. B. BIGGS		BLDG G560 DOMESTIC WATER LINE REPAIRS			
DR. B. BIGGS					
CHK. C J MAAS, PE					
SUBMITTED BY: B. BIGGS					
DESIGN DIR. C J MAAS, PE					
APPROVED: PWO OR OICC	DATE	SIZE	CODE IDENT. NO	NAVFAC DRAWING NO. 60042362	
		E1	80091	CONST. CONTR. N40085-24-0030	
SATISFACTORY TO:	DATE	SCALE: NOTED		SPEC. 05-24-0030 SHEET 2 OF 8	

REVISIONS			
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△	AMENDMENT 001	4/16/2026	

PLUMBING DEMO GENERAL NOTES

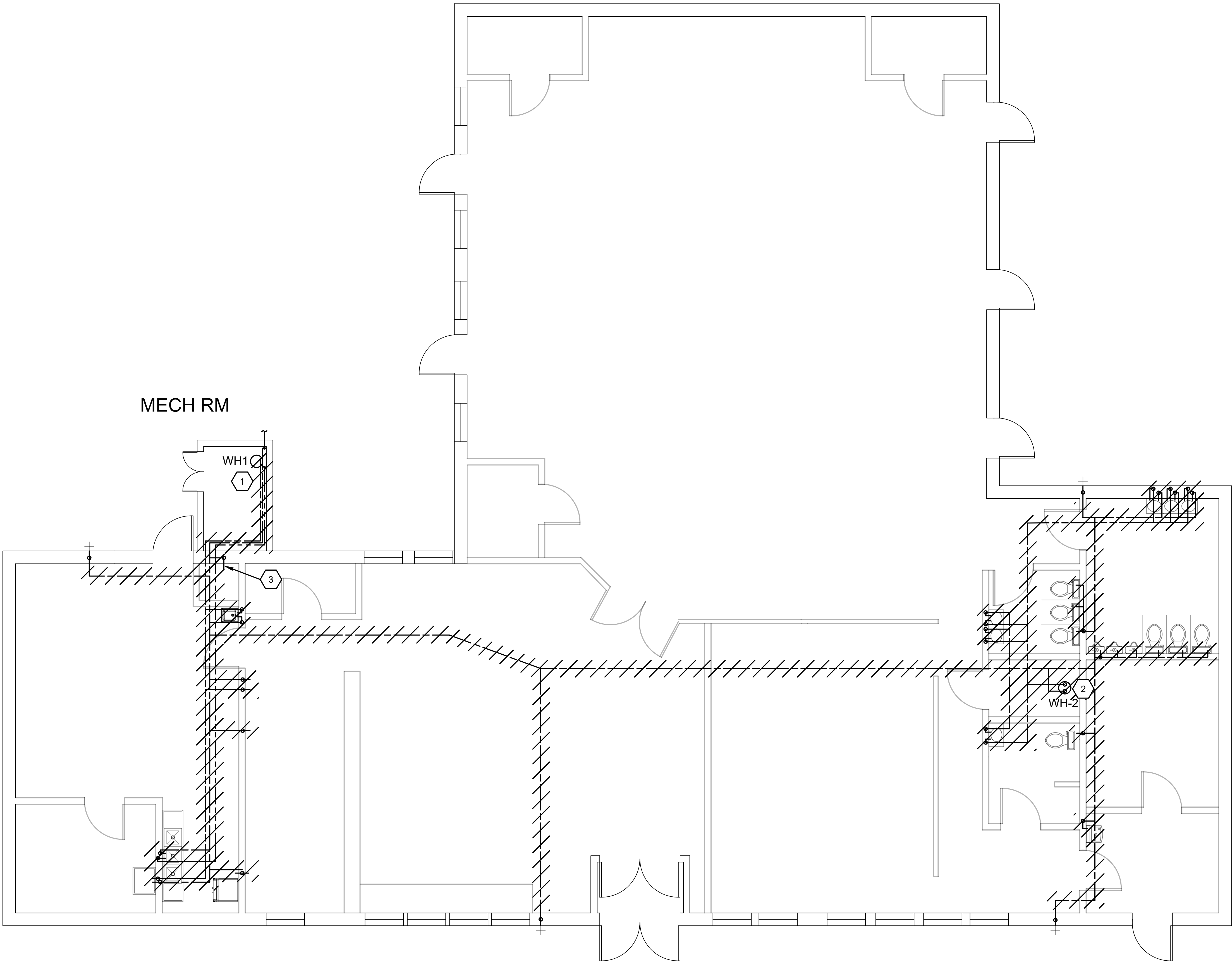
1. CONTRACTOR TO SCHEDULE WATER AND ELECTRICAL OUTAGES WITH THE CONTRACTING OFFICER REPRESENTATIVE
2. CONTRACTOR TO DEMOLISH ALL DOMESTIC WATER LINES INSIDE OF THE MECHANICAL ROOM AND BUILDING EXCEPT WHERE SHOWN TO REMAIN.
3. CONTRACTOR IS RESPONSIBLE FOR REPAIRING ANY DAMAGES TO THE BUILDING.
4. CONTRACTOR TO PRESERVE, PROTECT, AND REUSE ALL PLUMBING FIXTURES.

PLUMBING DEMO NOTES

- 1 DEMO WATER HEATER AND APPURTENANCES. DEMO ELECTRICAL FEEDING WATER HEATER AND RECIRCULATION PUMP IN ITS ENTIRETY BACK TO PANEL.
- 2 DEMO WATER HEATER AND APPURTENANCES. DEMO ELECTRICAL FEEDING WATER HEATER IN ITS ENTIRETY BACK TO PANEL.
- 3 DEMO STUB OUT COMPLETE. PATCH HOLE IN TILE.
- 4 EXISTING HOSE BIB TO REMAIN. EXISTING PIPE IN WALL CAVITY TO REMAIN.

NOTICE OF POTENTIAL LEAD-BASED PAINT AND PLUMBING MATERIALS

BUILDING G560 WAS ORIGINALLY CONSTRUCTED IN APPROXIMATELY 1977. IN ACCORDANCE WITH FEDERAL REGULATIONS, THE CONTRACTOR SHALL PRESUME THAT LEAD-BASED PAINT IS PRESENT IN ALL PAINTED SURFACES, BOTH INTERIOR AND EXTERIOR, UNLESS TESTING BY A CERTIFIED INSPECTOR PROVES OTHERWISE. THE CONTRACTOR SHALL PRESUME THAT PLUMBING PIPES, SOLDER AND OTHER COMPONENTS ALSO CONTAIN LEAD.



CONSTRUCTION DOCUMENTS				P101	
		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND			
		MARINE CORPS BASE CAMP LEJEUNE, NORTH CAROLINA			
DES. B. BIGGS		BLDG G560 DOMESTIC WATER LINE REPAIRS			
DR. B. BIGGS					
CHK. C J MAAS, PE					
SUBMITTED BY: B. BIGGS					
DESIGN DIR. C J MAAS, PE					
APPROVED: PWO OR OICC	DATE	SIZE	CODE IDENT. NO	NAVFAC DRAWING NO. 60042363	
		E1	80091	CONST. CONTR. N40085-24-0030	
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		SCALE: NOTED		SPEC. 05-24-0030 SHEET 3 OF 8	

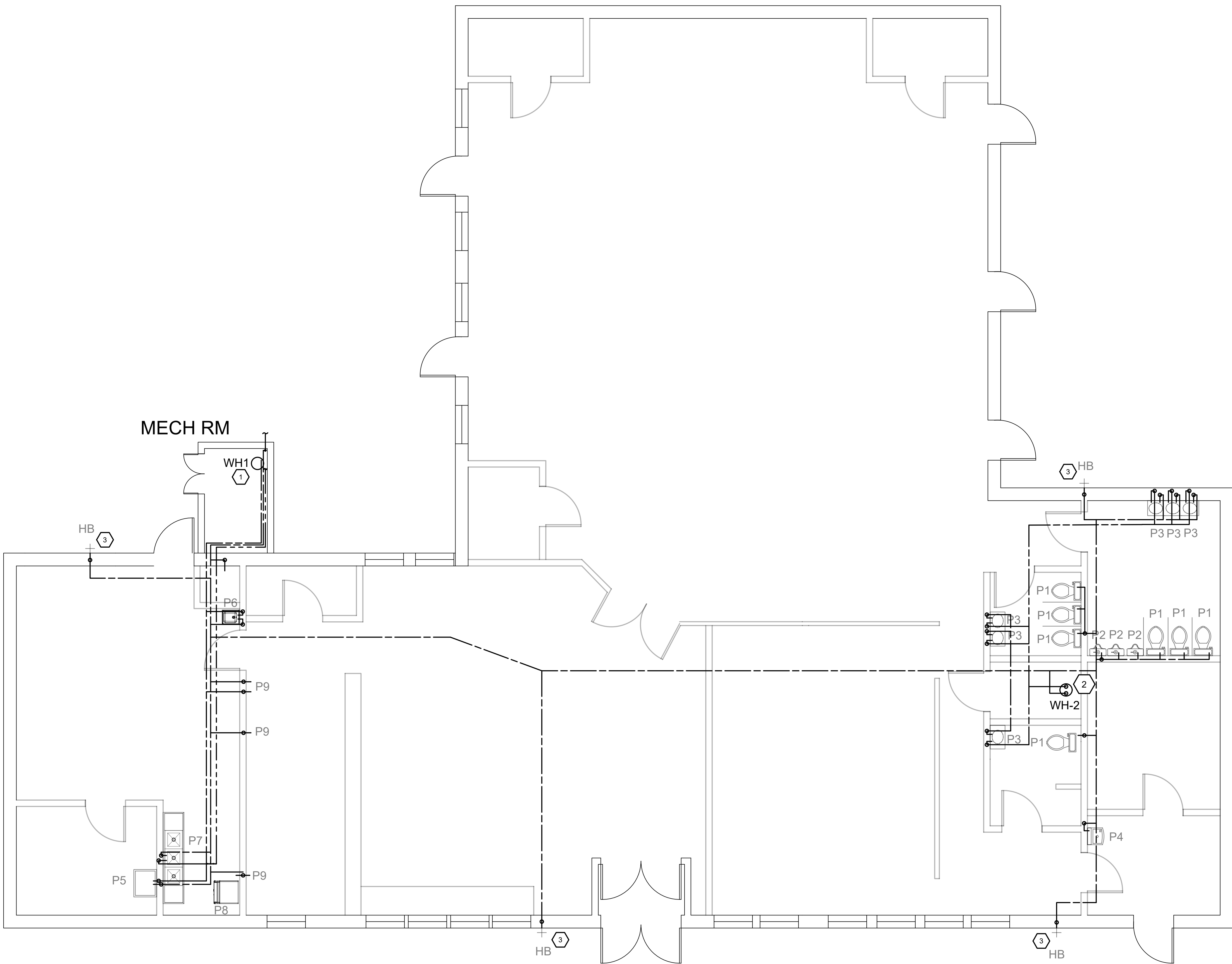
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SYM		DATE	APPROVED
Δ	AMENDMENT 001	4/16/2026	

PLUMBING GENERAL NOTES

1. PROVIDE NEW BALL-TYPE SHUTOFF VALVES AT EACH FIXTURE AND RECONNECT FIXTURES. WATER CLOSETS AND URINALS ARE TO ONLY RECEIVE A BRANCH SHUTOFF VALVE AS SHOWN IN P701.
2. TRAP PRIMERS SHALL BE REINSTALLED.
3. PROVIDE POINT-OF-USE MIXING VALVES TO PREVENT SCALDING AT ALL LAVATORIES AND HAND WASH SINKS.
4. ALL BALL VALVES, MIXING VALVES AND HAMMER ARRESTORS SHALL BE EASILY ACCESSIBLE. PROVIDE LABELED ACCESS HATCHES WHERE NECESSARY.
5. ALL WATER LINES SHALL BE INSULATED PER SPEC AND LABELED PER ASME A13.1
6. REPAIR, PATCH AND FINISH TO MATCH EXISTING CONDITIONS IN ALL DISTURBED AREAS.

PLUMBING KEYED NOTES

- 1 PROVIDE AND INSTALL NEW RPZ TYPE BACKFLOW PREVENTER, ELECTRIC WATER HEATER SET TO 140F, EXPANSION TANK, AND RECIRCULATING PUMP. REFER TO DETAILS A1 AND E1 ON P501.
- 2 PROVIDE AND INSTALL NEW 15 GALLON ELECTRIC WATER HEATER SET TO 140F AND EXPANSION TANK. REFER TO DETAIL D1 ON P501.
- 3 CONNECT TO EXISTING HOSE BIB PIPING ABOVE CEILING. PROVIDE INTERIOR-ACCESSIBLE BRANCH SHUTOFF BALL VALVE FOR EACH.



A1

FLOOR PLAN - WATER

3/16" = 1'-0"

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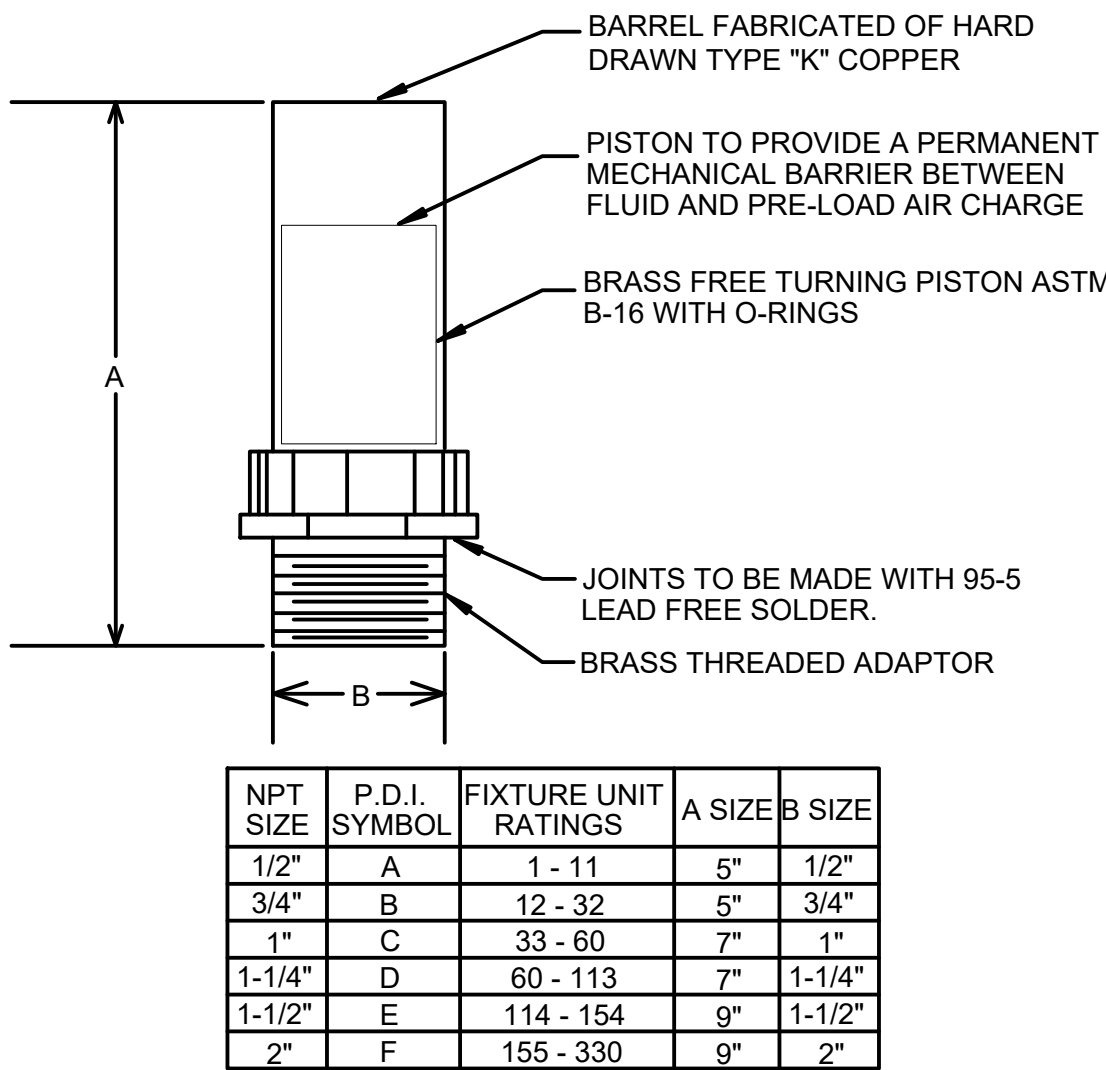
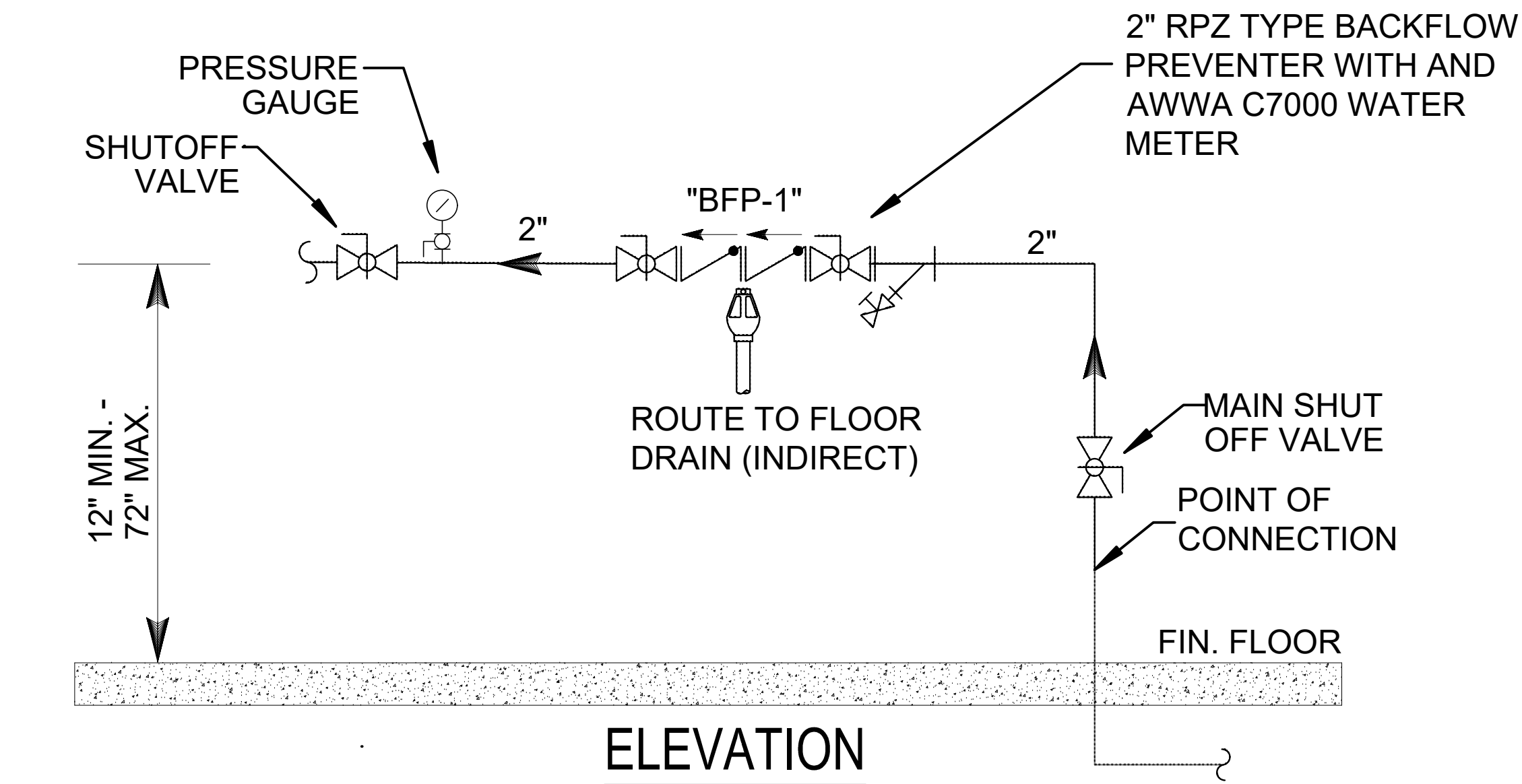
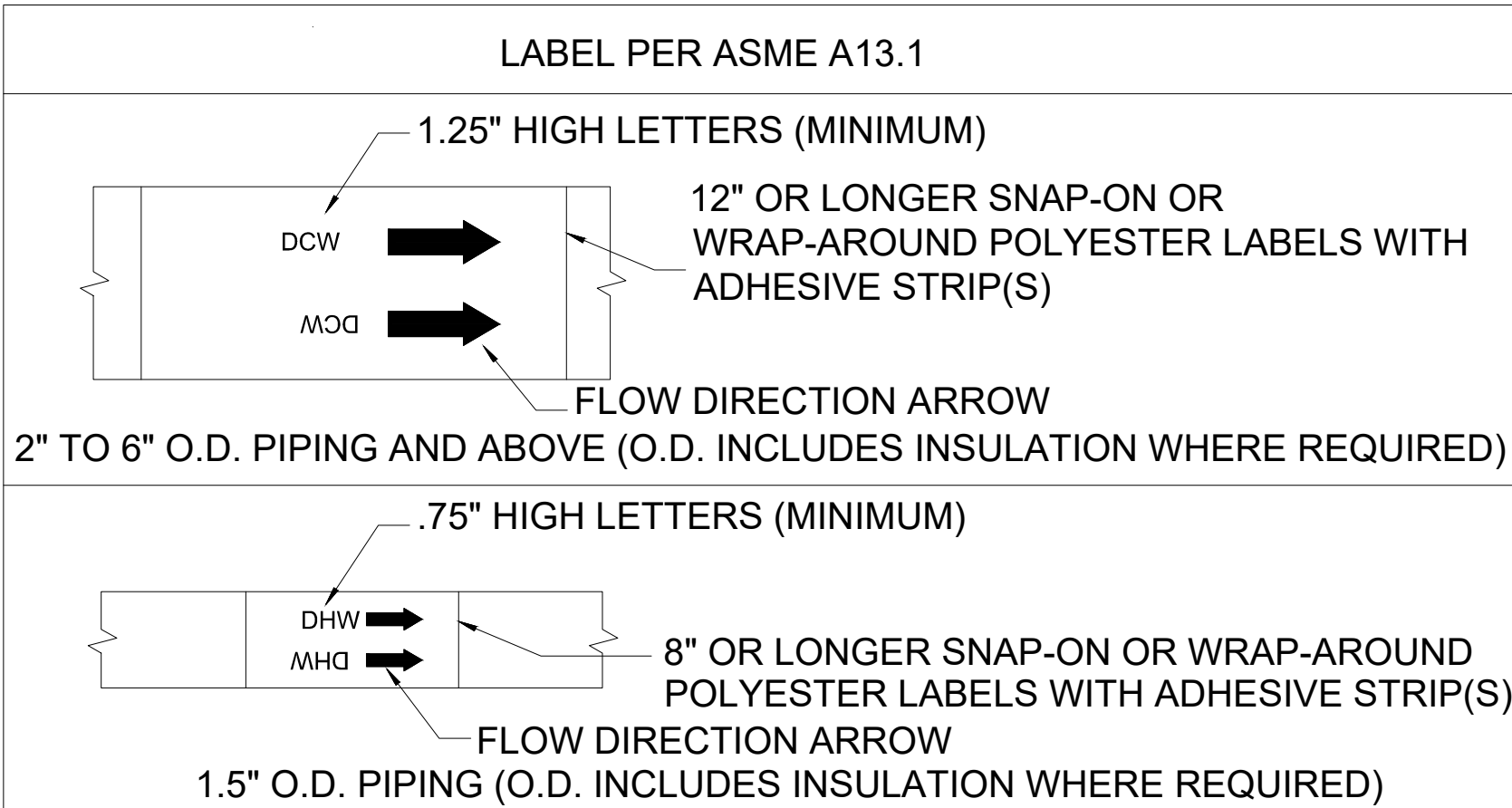
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PLAN NORTH

CONSTRUCTION DOCUMENTS			P102		
		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND			
		MARINE CORPS BASE CAMP LEJEUNE, NORTH CAROLINA			
DES. B. BIGGS		BLDG G560 DOMESTIC WATER LINE REPAIRS			
DR. B. BIGGS					
CHK. C J MAAS, PE					
SUBMITTED BY: B. BIGGS					
DESIGN DIR. C J MAAS, PE					
APPROVED: PWO OR OICC	DATE	SIZE	CODE IDENT. NO	NAVFAC DRAWING NO.	
		E1	80091	60042364	
		CONST. CONTR. N40085-24-0030			
SATISFACTORY TO:	DATE	SCALE: NOTED		SPEC. 05-24-0030	
				SHEET 4 OF 8	

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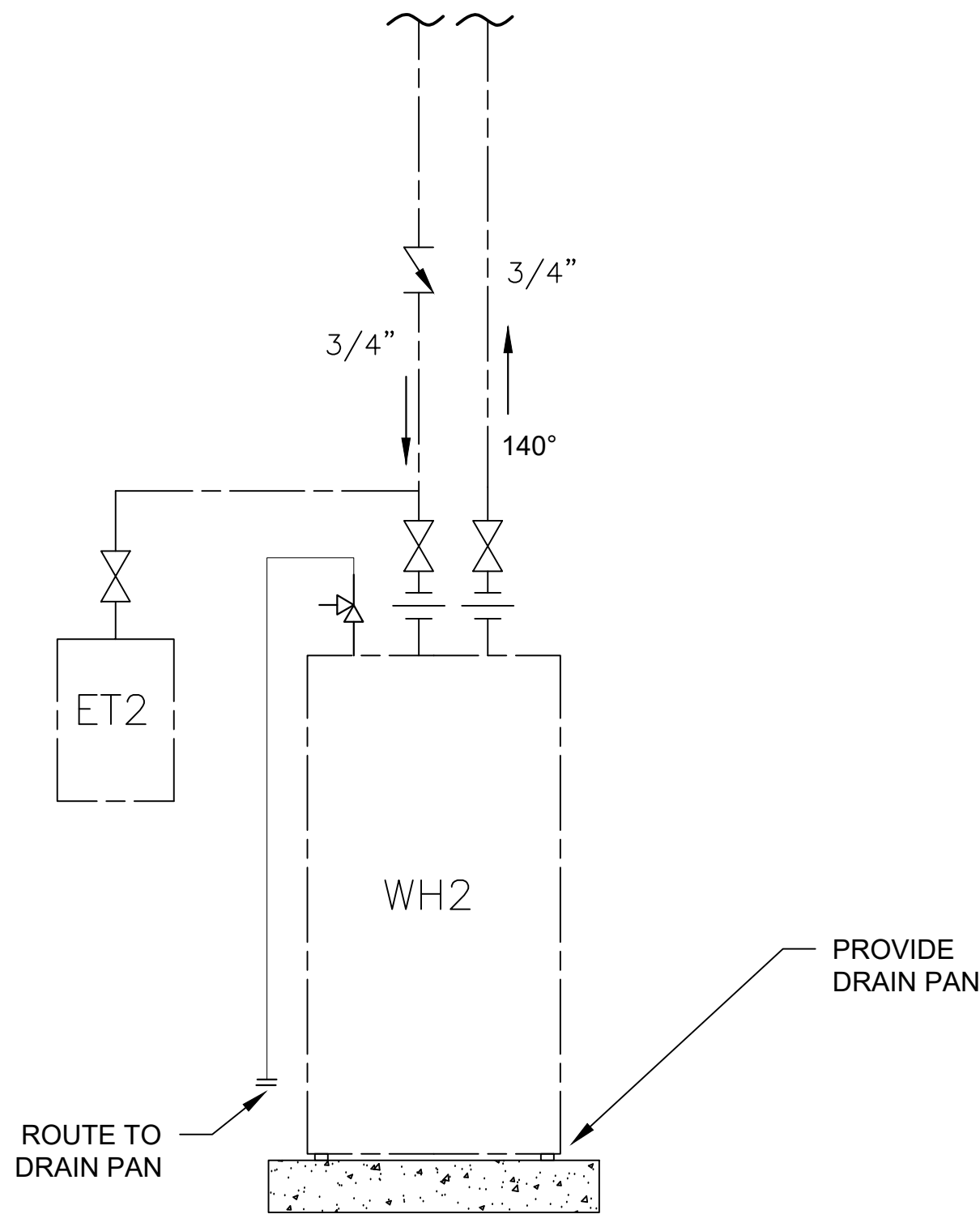
LABEL TEXT AND COLOR LEGEND		
PIPE SYSTEM DESCRIPTION	LETTER COLOR	BACKGROUND COLOR
COLD WATER SUPPLY	GREEN	WHITE
HOT WATER SUPPLY	GREEN	WHITE
HOT WATER RETURN	GREEN	WHITE



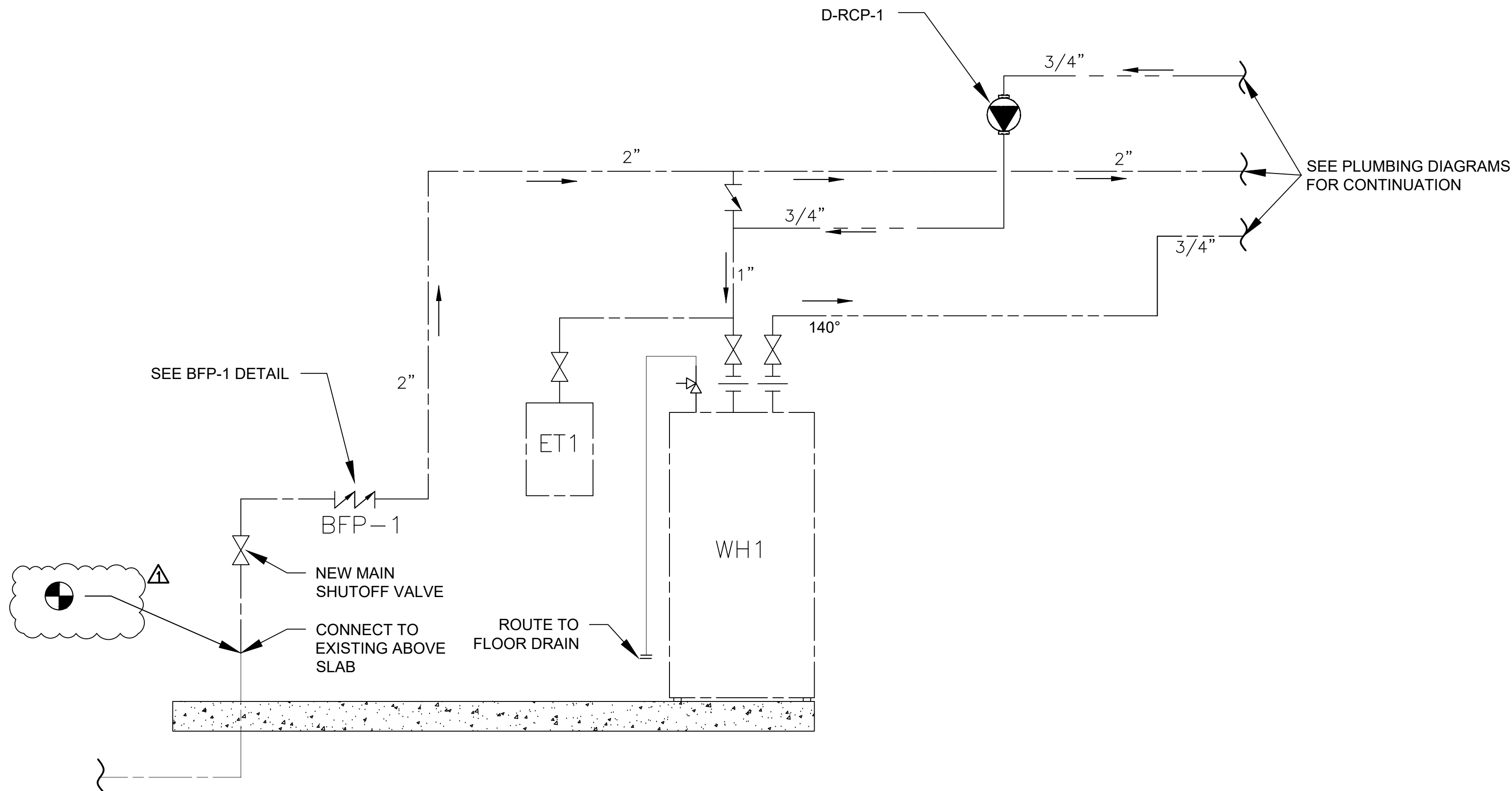
C1 PIPE LABELING DETAIL

B1 SHOCK ABSORBER DETAIL

A1 BACKFLOW PREVENTER DETAIL (BFP-1)



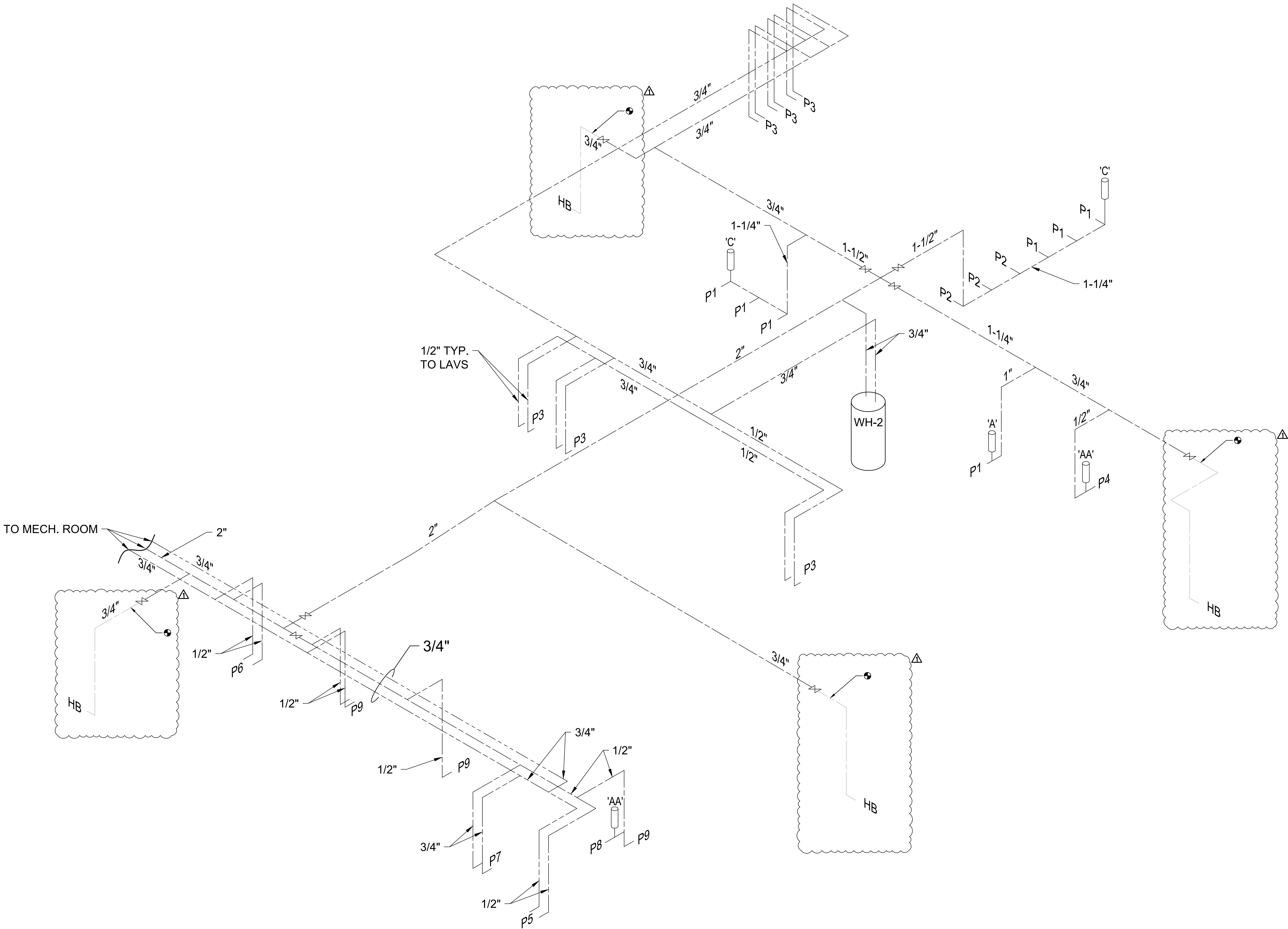
D1 WH-2 DETAIL



D1 WH-1 DETAIL

CONSTRUCTION DOCUMENTS		P501	
DEPARTMENT OF THE NAVY		NAVAL FACILITIES ENGINEERING COMMAND	
MARINE CORPS BASE		CAMP LEJEUNE, NORTH CAROLINA	
BLDG G560 DOMESTIC WATER LINE REPAIRS			
DES. B. BIGGS			
DR. B. BIGGS			
CHK. C J MAAS, PE			
SUBMITTED BY: B. BIGGS			
DESIGN DIR. C J MAAS, PE			
APPROVED: PWO OR OICC	DATE	SIZE	CODE IDENT. NO
		E1	80091
SATISFACTORY TO:	DATE	NAVFAC DRAWING NO.	CONST. CONTR.
		60042365	N40085-24-0030
		SCALE: NOTED	SPEC. 05-24-0030
			SHEET 5 OF 8

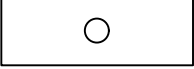
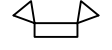
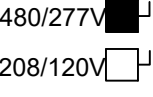
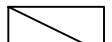
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A1 RISER DIAGRAM - WATER
NTS

CONSTRUCTION DOCUMENTS		P701	
DES. B. BIGGS		DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND	
DR. B. BIGGS		MARINE CORPS BASE	
CHK. C J MAAS, PE		CAMP LEJEUNE, NORTH CAROLINA	
SUBMITTED BY: B. BIGGS		BLDG G560 DOMESTIC	
DESIGN DIR. C J MAAS, PE		WATER LINE REPAIRS	
APPROVED: PWO OR OICC		DATE	SIZE
SATISFACTORY TO:		DATE	CODE IDENT. NO
		SCALE: NOTED	NAVFAC DRAWING NO.
			60042366
			CONST. CONTR. N40085-24-0030
			SHEET 6 OF 8

ELECTRICAL LEGEND

SYMBOL	DESCRIPTION
	LIGHT FIXTURE. SEE LIGHTING FIXTURE SCHEDULE.
	EMERGENCY LIGHTING UNIT, 2-HEAD WITH BATTERY BACK-UP. WALL MOUNTED, CONNECT TO NON-SWITCHED LEG. SEE DETAIL NL-26 & LUMINAIRE SCHEDULE.
	WALL SWITCH AT 46" AFF UNLESS OTHERWISE NOTED. LOW VOLTAGE. ASSOCIATED WIRING IS TO BE PROVIDED BUT IS NOT SHOWN ON DRAWINGS FOR CLARITY. PROVIDE SWITCHES COMPATIBLE FOR LUMAIRESLIGHTING SYSTEM. SEE OCCUPANCY SENSOR DIAGRAMS (E-108).
	RECEPTACLE, DUPLEX, 120VAC, 20A. UNLESS OTHERWISE NOTED. SEE MOUNTING HEIGHT DETAIL THIS SHEET. GFCI - GROUND FAULT CIRCUIT INTERRUPTER, 120VAC, 20A WP - WEATHERPROOF GFCI 120VAC,20A
	DISCONNECT SWITCH HEAVY DUTY, SIZE AS INDICATED ON DRAWINGS #NA = DISCONNECT SIZE / # = NUMBER OF POLES / # = NEMA RATING. /NAF = FUSE SIZE
	HOME RUN CIRCUITS IN CONDUIT BACK TO PANEL BOARD. SIZED AS INDICATED
	SWITCHED LEG NON-SWITCHED LEG
	PANELBOARD, SURFACE OR RECESSED MOUNTED AS SHOWN. SIZE, RATINGS, AND MOUNTING AS INDICATED ON PANEL SCHEDULE. CONTRACTOR IS RESPONSIBLE FOR REQUIRED CLEARANCE IN FRONT OF ELECTRICAL PANEL. SEE NEC TABLE 110.26 WORKING SPACES FOR ADDITIONAL CLEARANCE CONDITIONS.
	MECHANICAL EQUIPMENT; TYPE AND CONNECTION AS INDICATED

ABBREVIATIONS

A, AMP	AMPERE	LP	LIGHTING PANEL, LIGHT POLE
AFF	ABOVE FINISHED FLOOR	LTG	LIGHTING
AFG	ABOVE FINISHED GRADE	MCB	MAIN CIRCUIT BREAKER
AHU	AIR HANDLING UNIT	MCC	MOTOR CONTROL CENTER
AIC	AMPERE INTERRUPTING CAPACITY	MCP	MOTOR CIRCUIT PROTECTOR
ATS	AUTOMATIC TRANSFER SWITCH	MDP	MAIN DISTRIBUTION PANEL
AWG	AMERICAN WIRE GAUGE	MFR	MANUFACTURER
BOF	BOTTOM OF FIXTURE	MH	MANHOLE
BRKR	BREAKER	MLO	MAIN LUGS ONLY
C, CND	CONDUIT	MTD	MOUNTED
CAB	CABINET	MTG	MOUNTING
CAT	CATALOG	MTS	MANUAL TRANSFER SWITCH
CKT	CIRCUIT	MV	MEDIUM VOLTAGE
CB	CIRCUIT BREAKER	N / NEUT	NEUTRAL
CCTV	CLOSED CIRCUIT TELEVISION	NA	NOT APPLICABLE
CKT	CIRCUIT	NC	NORMALLY CLOSED
CLG	CEILING	NEC	NATIONAL ELECTRIC CODE
CP	CONTROL PANEL	NIC	NOT IN CONTRACT
CR	CONTROL RELAY, CORROSION RESISTANT	NL	NIGHT LIGHT
CS	CONTROL SWITCH	NO	NORMALLY OPEN
CV	CONTROL VALVE	NTS	NOT TO SCALE
CT	CURRENT TRANSFORMER	P	POLE
CU	COPPER	PA	PUBLIC ADDRESS
EF	EXHAUST FAN	PB	PULL BOX, PUSH-BUTTON
EMER	EMERGENCY	PF	POWER FACTOR
EMT	ELECTRICAL METALLIC TUBING	PH	PHASE
ENCL	ENCLOSURE	PLC	PROGRAMMABLE LOGIC CONTROLLER
EQUIP	EQUIPMENT	PNL	PANEL
EWIC	ELECTRIC WATER COOLER	P	POWER PANEL, POWER POLE
EWI	ELECTRIC WATER HEATER	PPE	PORTABLE POWER EQUIPMENT CABLE
EPRF	EXPLOSION PROOF	PT	POTENTIAL TRANSFORMER
FA	FIRE ALARM	PWR	POWER
FAAP	FIRE ALARM ANNUNCIATOR PANEL	RECPT, RCP	RECEPTACLE
FACP	FIRE ALARM CONTROL PANEL	REQD	REQUIRED
FBD	FURNISHED BY OTHERS	RGS	RIGID GALVANIZED STEEL CONDUIT
FLA	FULL LOAD AMPS	RM	ROOM
FLUOR	FLUORESCENT	RTU	REMOTE TELEMTRY UNIT
FLR	FLOOR	DCM	DC MOTOR DRIVE
FWE	FURNISHED WITH EQUIPMENT	SH	SHEET
GEN	GENERATOR	SPD	SURGE PROTECTION DEVICE
G, GND	GROUND	SPEC	SPECIFICATION
GFCI	GROUND FAULT CURRENT INTERRUPTER	SS	SELECTOR SWITCH
HH	HANDHOLE	SST	STAINLESS STEEL
HID	HIGH INTENSITY DISCHARGE	SW	SWITCH
HOA	HAND-OFF-AUTO	SWBD	SWITCHBOARD
HP	HORSE POWER	SWGR	SWITCH GEAR
HPF	HIGH POWER FACTOR	TEL	TELEPHONE
HPS	HIGH PRESSURE SODIUM	TPS	TWISTED PAIR SHIELDED
HTR	HEATER	TVSS	TRANSIENT VOLTAGE SURGE SUPPRESSER
HV	HIGH VOLTAGE	TYP	TYPICAL
HZ	HERTZ	UGND	UNDERGROUND
IMC	INTERMEDIATE METALLIC CONDUIT	UH	UNIT HEATER
INCAND	INCANDESCENT	UON	UNLESS OTHERWISE NOTED
JB	JUNCTION BOX	UTIL	UTILITY
K	THOUSAND	V	VOLTS
KCMIL	THOUSAND CIRCULAR MILLS	VFD	VARIABLE FREQUENCY DRIVE
KVA	KILOVOLT AMPERE	W	WIRE, WATT
KW	KILOWATTS	WH	WATT-HOUR
KWH	KILOWATT-HOURS	WP	WEATHERPROOF
		WTE	WEATHERPROOF TELEPHONE ENCLOSURE
		XFMR	TRANSFORMER
		X/R	EXISTING RELOCATED

ELECTRICAL GENERAL NOTES:

- ALL ELECTRICAL WORK MUST BE IN FULL COMPLIANCE WITH NFPA 70 AND ALL LOCAL CODES AND ORDINANCES AND IN ACCORDANCE WITH THE REQUIREMENTS OF THE LOCAL AUTHORITY HAVING JURISDICTION.
- ALL EQUIPMENT PROVIDED BY THE CONTRACTOR MUST BE LISTED AND LABELED BY A NATIONALLY-RECOGNIZED TESTING AGENCY, ACCEPTABLE TO THE AUTHORITY HAVING JURISDICTION, FOR THE CONDITIONS OF INSTALLATION. ALL MATERIAL, EQUIPMENT AND DEVICES MUST BE NEW CURRENT PRODUCTS OF MANUFACTURERS REGULARLY ENGAGED IN THE PRODUCTION OF SUCH PRODUCTS. EQUIPMENT MUST BE SUITABLE FOR ITS APPLICATION (E.G. WHEN INSTALLED OUTDOORS, IT MUST BE WEATHERPROOF, ETC.)
- THE CONTRACTOR MUST REVIEW ALL DRAWINGS AND SPECIFICATIONS FOR WORK REQUIREMENTS, THE AMOUNT OF SPACE AVAILABLE FOR ELECTRICAL EQUIPMENT, AND LAYOUT HIS WORK IN A COMPATIBLE AND FUNCTIONAL MANNER.
- UNLESS SPECIFICALLY NOTED OTHERWISE, SYSTEMS PROVIDED OR INSTALLED BY THE CONTRACTOR MUST BE COMPLETE AND FULLY-FUNCTIONING AFTER INSTALLATION. INCIDENTAL COMPONENTS MAY NOT BE SHOWN, AND ALL WORK WHICH MAY BE REASONABLY IMPLIED AS BEING INCIDENTAL TO THIS WORK, BUT REQUIRED FOR THE PROPER OPERATION OF THE EQUIPMENT OR SYSTEM MUST BE PROVIDED BY THE CONTRACTOR AND INCLUDED IN THE BID. ADDITIONAL CIRCUITS MUST BE INSTALLED WHEREVER NEEDED TO CONFORM TO THE SPECIFIC REQUIREMENTS OF EQUIPMENT.
- TEMPORARY POWER CONNECTIONS AS REQUIRED MUST BE PROVIDED BY THE CONTRACTOR AND INCLUDED IN THE BID. ALL TEMPORARY EQUIPMENT WIRING MUST BE IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE. THE CONTRACTOR MUST PROVIDE DETAILS, METHODS, MATERIALS, ETC. FOR REVIEW PRIOR TO MAKING TEMPORARY CONNECTIONS. FURNISH AND INSTALL ALL EQUIPMENT AND MATERIALS INCLUDING CONTROL EQUIPMENT, MOTOR STARTERS, BRANCH AND FEEDER CIRCUIT BREAKERS, PANELBOARDS, TRANSFORMERS, ETC. FOR TEMPORARY POWER. COORDINATE WITH THE UTILITY PROVIDER AS REQUIRED.
- THE WORK MUST INCLUDE COMPLETE TESTING OF ALL EQUIPMENT AND WIRING AT THE COMPLETION OF WORK AND ANY MINOR CORRECTIONS, CHANGES OR ADJUSTMENTS NECESSARY FOR THE PROPER FUNCTIONING OF THE SYSTEM AND EQUIPMENT.
- ALL ELECTRICAL EQUIPMENT MUST, AT ALL TIMES DURING CONSTRUCTION, BE ADEQUATELY PROTECTED AGAINST MECHANICAL INJURY, OR DAMAGE BY WATER AND/OR THE ELEMENTS. ELECTRICAL EQUIPMENT MUST NOT BE STORED OUT OF DOORS, BUT MUST BE STORED IN DRY PERMANENT SHELTERS. IF AN APPARATUS HAS BEEN DAMAGED, OR HAS BEEN SUBJECT TO POSSIBLE INJURY BY WATER OR THE ELEMENTS, SUCH DAMAGE MUST BE REPLACED AT NO ADDITIONAL COST.
- DO NOT SCALE ELECTRICAL DRAWINGS. CONTRACTOR MUST FIELD VERIFY ALL DIMENSIONS.
- CIRCUIT LAYOUTS ARE NOT INTENDED TO SHOW THE NUMBER OF FITTINGS, OR OTHER INSTALLATION DETAILS. UNLESS NOTED OTHERWISE, THE EXACT ROUTING OF FEEDER AND BRANCH CIRCUIT RACEWAYS AND CABLES IS THE RESPONSIBILITY OF THE CONTRACTOR. RISER AND GENERAL CIRCUIT ARRANGEMENTS ARE SHOWN SCHEMATICALLY/DIAGRAMMATICALLY ONLY. THE CONTRACTOR MUST ROUTE CONDUITS AS REQUIRED BY THE CONDITIONS OF THE INSTALLATION.
- UNLESS DIMENSIONED, DEVICE LOCATIONS SHOWN ON THE DRAWINGS ARE APPROXIMATE. ADJUST EXACT LOCATIONS AS REQUIRED TO SERVE THE INTENDED PURPOSE AND TO AVOID CONFLICTS AND INTERFERENCES WITH OTHER TRADES. IF NOT SHOWN ON THE ARCHITECTURAL DRAWINGS OR DIMENSIONED ON THE ELECTRICAL DRAWINGS, VERIFY EXACT LOCATION WITH THE GOVERNMENT PRIOR TO ROUGH-IN.
- CONDUIT TERMINATING IN PRESSED STEEL BOXES MUST HAVE DOUBLE LOCKNUTS AND INSULATED BUSHINGS. CONDUITS TERMINATING IN GASKETED ENCLOSURES MUST BE TERMINATED WITH GROUNDING TYPE CONDUIT HUBS.
- BRANCH CIRCUIT HOMERUNS SHOWN ON DRAWINGS INDICATE PHASE. CONDUCTORS, NEUTRAL, EQUIPMENT GROUND CONDUCTORS AS REQUIRED. ADDITIONAL CONDUCTORS REQUIRED FOR CONTROL MUST BE INCLUDED EVEN IF NOT EXPLICITLY SHOWN.
- SEAL ALL CONDUIT OPENINGS THROUGH EXTERIOR BUILDING WALLS AND ROOF WATERTIGHT.
- IN WET LOCATIONS AND EXTERIOR, ALL WIRING DEVICES MUST BE WEATHER-RESISTANT LISTED WITH WEATHERPROOF WHILE IN USE COVER.
- RACEWAYS PENETRATING FLOORS, CEILINGS OR WALLS MUST BE PROPERLY SEALED SMOKE TIGHT.
- ALL RACEWAYS MUST BE CONCEALED WHERE POSSIBLE.
- INSTALL EXPOSED RACEWAYS PARALLEL TO OR AT RIGHT ANGLES TO NEARBY SURFACES OR STRUCTURAL MEMBERS, AND FOLLOW THE SURFACE CONTOURS AS MUCH AS POSSIBLE. NO DIAGONAL RUNS WILL BE ALLOWED. ALL CONDUITS MUST BE RUN STRAIGHT WITH QUALITY WORKMANSHIP-LIKE MANNER. RUN PARALLEL OR BANKED RACEWAYS TOGETHER ON COMMON SUPPORTS WHERE PRACTICAL. MAKE BENDS IN PARALLEL OR BANKED RUNS FROM SAME CENTERLINE TO MAKE BENDS PARALLEL.
- PATCHING OF WATERPROOFED SURFACES MUST RENDER THE AREA OF THE PATCHING COMPLETELY WATERPROOF.
- ALL MOTORS AND OTHER VIBRATING EQUIPMENT MUST BE CONNECTED TO THE CONDUIT SYSTEM BY MEANS OF A SHORT SECTION (18 INCH MINIMUM) OF FLEXIBLE CONDUIT UNLESS OTHERWISE INDICATED. AN EQUIPMENT GROUNDING CONDUCTOR MUST BE INSTALLED INSIDE THE FLEXIBLE CONDUIT AND TERMINATE AT THE LOAD END WITH AN APPROVED GROUNDING CLAMP OR LUG.
- SURFACE MOUNTED PANELBOARDS, JUNCTION, OUTLET AND PULL BOXES, RACEWAYS, ETC., INSTALLED ON EXTERIOR SURFACES OR INSIDE ON EXTERIOR WALLS MUST BE SUPPORTED BY SPACERS TO PROVIDE A MINIMUM 1/4" MINIMUM CLEARANCE BETWEEN THE WALL AND EQUIPMENT.
- TYPED PANELBOARD DIRECTORIES INSTALLED IN THE PANELBOARD DOOR POCKET MUST REFLECT FINAL CONDITIONS AND ACTUAL ROOM NAMES AND NUMBERS IN ADDITION TO THE GENERAL DESCRIPTION SHOWN ON THE PANEL SCHEDULES ON THE DRAWINGS.
- DO NOT PULL CONDUCTORS UNTIL THE CONDUIT SYSTEM IS COMPLETE IN EVERY DETAIL. IN THE CASE OF CONCEALED WORK, "COMPLETE" MEANS UNTIL ALL ROUGH PLASTERING OR MASONRY HAS BEEN COMPLETED.
- WHERE SIZE IS NOT SHOWN ON THE DRAWINGS, BRANCH CIRCUITS MUST CONSIST OF #12 OR #10 AWG MINIMUM PHASE, NEUTRAL AND EQUIPMENT GROUND CONDUCTORS IN 3/4" MINIMUM RACEWAY.
- USE #10 AWG CONDUCTORS FOR 20 AMPERE, 120 VOLT BRANCH CIRCUITS WITH A TOTAL INSTALLED LENGTH GREATER THAN 75 FEET AND/OR BRANCH CIRCUIT HOMERUNS LONGER THAN 50 FEET, I.E.; #12 AWG INCREASED TO #10 AWG FOR RECEPTACLE BRANCH CIRCUITS OVER 75 FEET TOTAL LENGTH (INCLUDING THE HOMERUN SEGMENT) AND HOMERUNS OVER 50 FEET. IF 277 VOLT CIRCUITS ARE SHOWN, USE #10 AWG CONDUCTORS FOR 20 AMPERE, 277 VOLT BRANCH CIRCUITS WITH TOTAL INSTALLED LENGTH GREATER THAN 200 FEET AND/OR BRANCH CIRCUIT HOMERUNS LONGER THAN 125 FEET, I.E.; #12 AWG INCREASED TO #10 AWG FOR RECEPTACLE BRANCH CIRCUITS OVER 75 FEET TOTAL LENGTH (INCLUDING THE HOMERUN SEGMENT) AND HOMERUNS OVER 50 FEET.
- KEEP CONDUCTOR SPLICES TO A MINIMUM. INSTALL SPLICES AND TAPES THAT POSSESS EQUIVALENT OR BETTER MECHANICAL STRENGTH AND INSULATION RATINGS THAN CONDUCTORS BEING SPLICED. USE SPLICE AND TAP CONNECTORS COMPATIBLE WITH CONDUCTOR MATERIAL. INSTALL CONDUCTORS AT EACH OUTLET WITH AT LEAST 6 INCHES OF SLACK. CONNECT OUTLETS AND COMPONENTS TO WIRING AND TO GROUND AS INDICATED AND INSTRUCTED BY THE MANUFACTURER.
- DO NOT SPLICE BRANCH CIRCUIT HOMERUNS WITHOUT THE PERMISSION OF THE GOVERNMENT. HOMERUNS MUST BE CONTINUOUS FROM THE LAST OUTLET BOX TO THE SERVING PANELBOARD.
- DO NOT COMBINE BRANCH CIRCUIT HOMERUNS UNLESS SPECIFICALLY INDICATED ON THE DRAWINGS.
- INSTALL WIRING DEVICES AT HEIGHTS AS SHOWN ON THE DRAWINGS.
- PROVIDE GROUND FAULT CIRCUIT-INTERRUPTER PROTECTION FOR PERSONNEL IN ACCORDANCE WITH THE NEC INCLUDING EXTERIOR RECEPTACLES AND RECEPTACLES IN AREAS SUBJECT TO POSSIBLE WET CONDITIONS.
- COORDINATE LOCATIONS OF MECHANICAL EQUIPMENT WITH THE RESPECTIVE CONTRACTOR, VENDORS AND THE GOVERNMENT BEFORE ROUGH-IN. ADJUST LIGHTING FIXTURES, RECEPTACLES AND ELECTRICAL EQUIPMENT TO ACCOMMODATE THIS EQUIPMENT. ADVISE THE GOVERNMENT OF CONFLICTS BEFORE ROUGH-IN.
- BEFORE COMMENCING WORK OR ORDERING MATERIALS, THE CONTRACTOR MUST COORDINATE WITH OTHER TRADES AND VERIFY THE NAMEPLATE RATINGS OF ALL EQUIPMENT (MOTORS, HEATERS, COMPRESSORS, ETC.) AND ADJUST THE RATINGS OF THE ELECTRICAL EQUIPMENT (SWITCHES, FUSES, CIRCUIT BREAKERS, FEEDERS, ETC.) AS APPROPRIATE TO SERVE THIS EQUIPMENT.
- ENERGIZE EQUIPMENT ONLY AFTER OBTAINING PERMISSION FROM THE CONTRACTOR PROVIDING THE EQUIPMENT.
- UNLESS SPECIFICALLY NOTED OTHERWISE, THE CONTRACTOR MUST MAKE FINAL CONNECTIONS TO ALL UTILIZATION EQUIPMENT SHOWN ON THE DRAWINGS. VERIFY THE TYPE OF FINAL CONNECTION AND PROVIDE APPROPRIATE WIRING METHOD. THE CONTRACTOR MUST COORDINATE WITH THE MECHANICAL TRADE, PRIOR TO ORDERING OR INSTALLATION OF ANY EQUIPMENT. TO VERIFY MECHANICAL EQUIPMENT REQUIREMENTS ARE PROVIDED IN THE ELECTRICAL DESIGN. THE CONTRACTOR WILL NOT BE COMPENSATED FOR COSTS ASSOCIATED WITH CHANGING THE ELECTRICAL SYSTEMS TO MATCH UTILIZATION EQUIPMENT, EVEN IF THE ELECTRICAL WORK IS INSTALLED PER THE ELECTRICAL DRAWINGS.
- THE CONTRACTOR MUST COORDINATE ALL EQUIPMENT TERMINATIONS, PLUGS AND CORDSETS WITH VENDOR EQUIPMENT AND VERIFY ALL DEVICE LOCATIONS FOR SPECIALITY EQUIPMENT WITH CASEWORK PRIOR TO ROUGH-IN.
- INSTALLATION INFORMATION PACKED WITH DEVICES AND EQUIPMENT MUST BE RETAINED FOR INCLUSION IN THE OPERATIONS AND MAINTENANCE MANUALS.
- THE CONTRACTOR SHALL VISIT THE SITE AND BECOME FAMILIAR WITH THE EXISTING ELECTRICAL SYSTEMS AND THE EXISTING BUILDING. THE SUBMISSION OF THE PROPOSAL BY THE CONTRACTOR SHALL BE CONSIDERED EVIDENCE THAT HE OR HIS REPRESENTATIVE HAS VISITED THE SITE AND BUILDINGS AND NOTED THE LOCATION AND CONDITIONS UNDER WHICH THE WORK WILL BE PERFORMED AND THAT HE TAKES FULL RESPONSIBILITY OF ALL FACTORS GOVERNING HIS WORK. NO EXTRAS WILL BE CONSIDERED BECAUSE OF ADDITIONAL WORK NECESSITATED BY EXISTING JOB CONDITIONS THAT ARE NOT INDICATED ON THE DRAWINGS.
- CONTRACTOR SHALL VERIFY ALL EQUIPMENT AND MECHANICAL CHARACTERISTICS PRIOR TO BEGINNING WORK. CONTRACTOR SHALL MATCH ELECTRICAL REQUIREMENTS AS REQUIRED BY MECHANICAL EQUIPMENT.
- SAFETY
 - COMPLY WITH OSHA AND NEC ARC FLASH PROTECTION REQUIREMENTS.
 - FOR EQUIPMENT BEING REMOVED AND REPLACED, THE CONTRACTOR SHALL DE-ENERGIZE THE EQUIPMENT AND MAKE IT SAFE PRIOR TO REMOVAL AND COMPLY WITH OSHA REQUIREMENTS FOR LOCKING-OUT AND TAGGING EQUIPMENT TO PREVENT INADVERTENT RE-ENERGIZING.

REVISIONS		
SYM	DATE	APPROVED

CONSTRUCTION DOCUMENTS

E001

DES. B. BIGGS DR. B. BIGGS CHK. C J MAAS, PE SUBMITTED BY: B. BIGGS DESIGN DIR. C J MAAS, PE	DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND MARINE CORPS BASE CAMP LEJEUNE, NORTH CAROLINA		
	BLDG G560 DOMESTIC WATER LINE REPAIRS		
	APPROVED: PWO OR OICC	DATE	NAVFAC DRAWING NO. 60042367
	SATISFACTORY TO:	DATE	CONST. CONTR. N40085-24-0030
SCALE: NOTED		SPEC. 05-24-0030	
		SHEET 7 OF 8	

REVISIONS			
SYM		DATE	APPROVED

ELECTRICAL SHEET NOTES

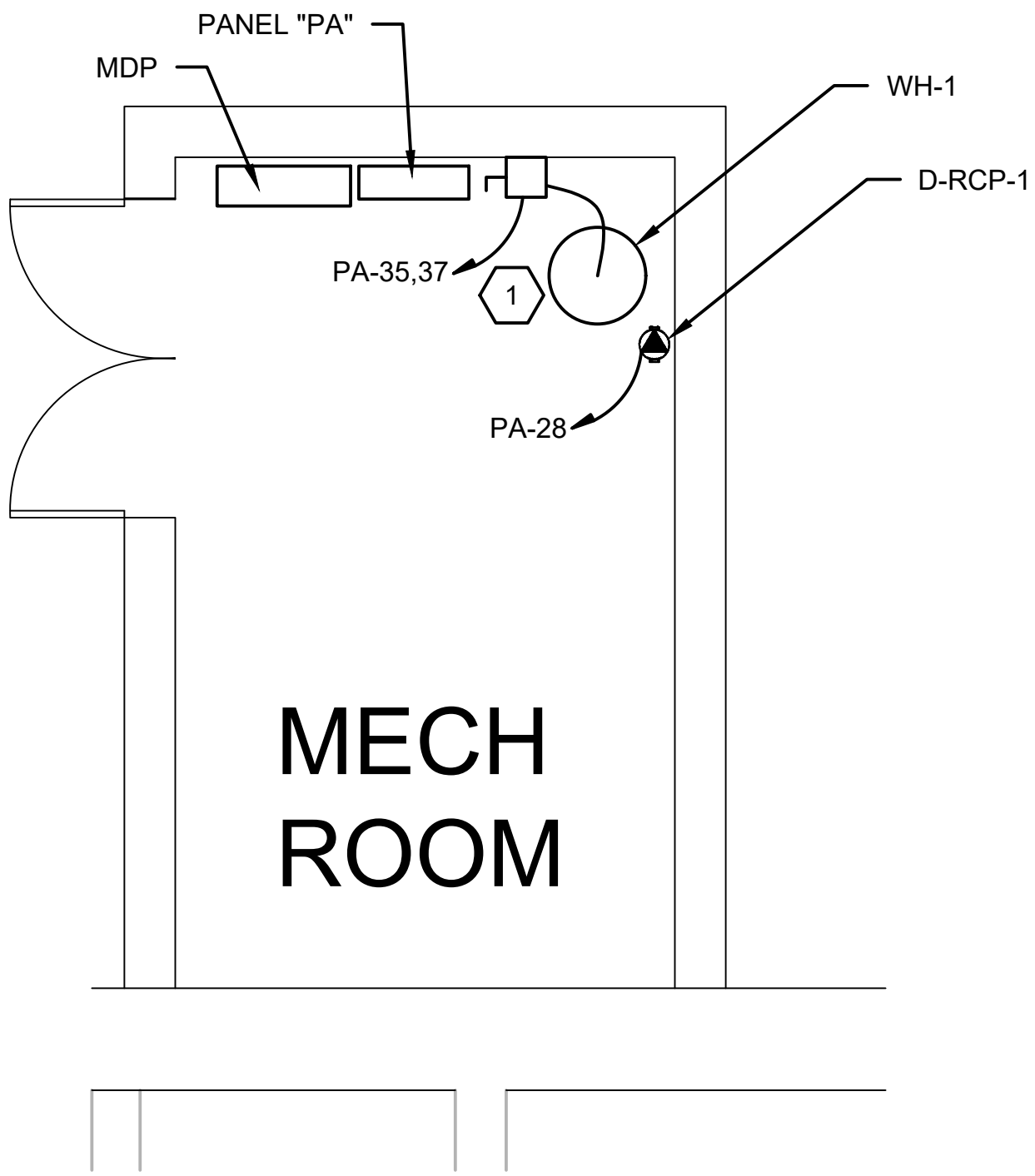
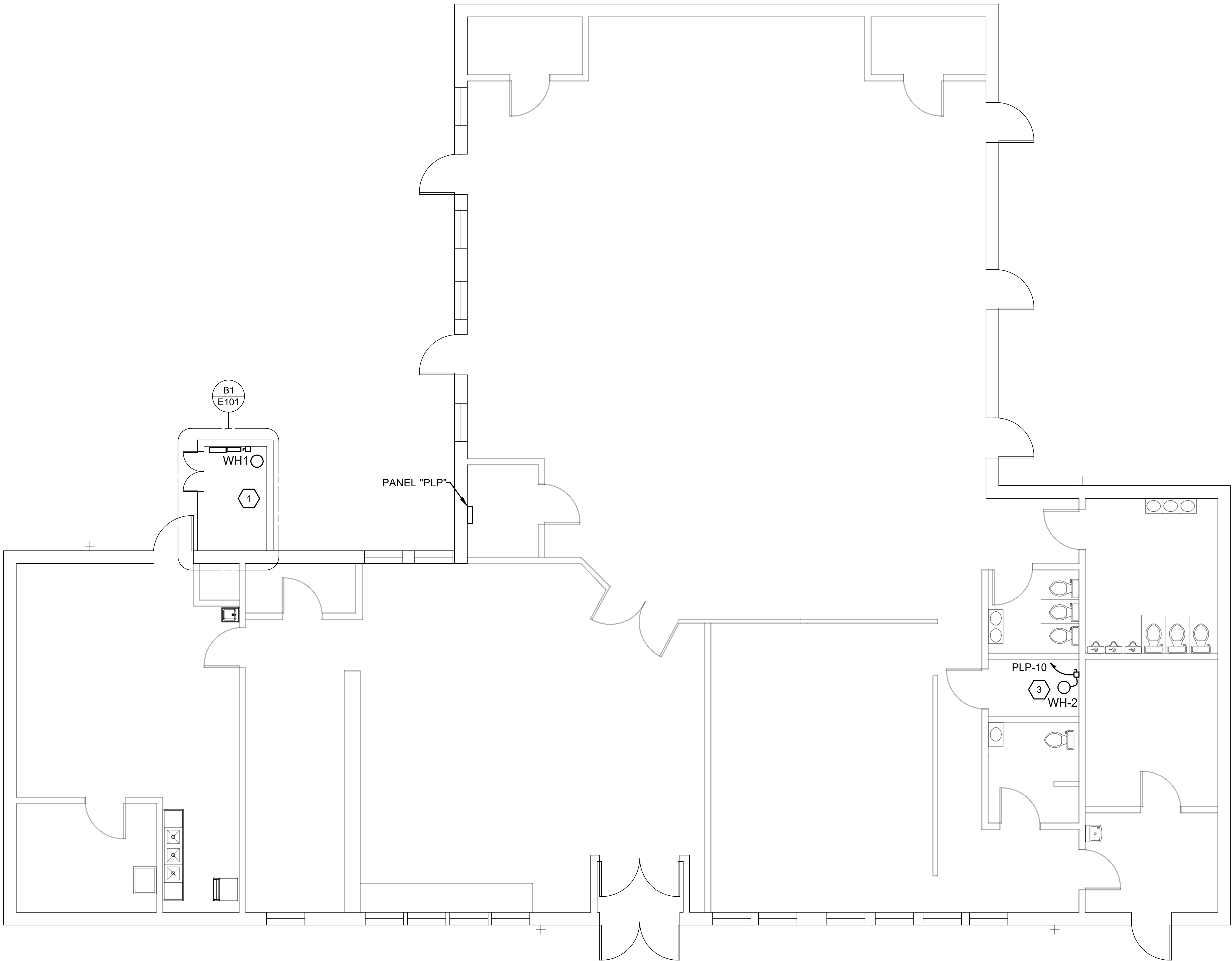
- ELECTRICAL CONDUITS, CONDUCTORS, DISCONNECT AND BREAKERS SERVING EXISTING WATER HEATERS AND RECIRCULATING PUMP SHALL BE DEMOLISHED IN ITS ENTIRETY.
- PROVIDE AND INSTALL ALL NEW CONDUIT, CONDUCTORS, BREAKERS AND DISCONNECTS.

ELECTRICAL KEYED NOTES

- 1 PROVIDE AND INSTALL ELECTRIC WATER HEATER AND DISCONNECT.
- 2 PROVIDE AND INSTALL NEW RECIRCULATING PUMP.
- 3 PROVIDE AND INSTALL ELECTRIC WATER HEATER AND DISCONNECT.

NEW ELECTRICAL SCHEDULE

MARK	PNL NAME	BREAKER LOCATION	BKR SIZE	DISCONNECT	CONDUCTOR QUANTITY AND SIZE	CONDUIT SIZE	ELECTRICAL (VOLTS / PH)
WH-1	PA	35,37	40A / 2	60A/3R	2 - #8, 1 - #8 EGC	1"	208 / 1
WH-2	PLP	10	30A / 1	30A/3R	2 - #10, 1 - #10 EGC	3/4"	120 / 1
D-RCP-1	PA	28	20A / 1	N/A	2 - #12, 1 - #12 EGC	1/2"	120 / 1



B1 MECHANICAL ROOM DETAIL - ELECTRICAL
NTS

A1 FLOOR PLAN - ELECTRICAL

3/16" = 1'-0"

0 4 8 12

PLAN NORTH

CONSTRUCTION DOCUMENTS

E101

DES. B. BIGGS		NAVAL FACILITIES ENGINEERING COMMAND	
DR. B. BIGGS		MARINE CORPS BASE	
CHK. C J MAAS, PE		CAMP LEJEUNE, NORTH CAROLINA	
SUBMITTED BY: B. BIGGS		BLDG G560 DOMESTIC WATER LINE REPAIRS	
DESIGN DIR. C J MAAS, PE		NAVFAC DRAWING NO. 60042367A	
APPROVED: PWO OR OICC		SIZE E 1	CODE IDENT. NO 80091
SATISFACTORY TO:		DATE	CONST. CONTR. N40085-24-0030
		SCALE: NOTED	SPEC. 05-24-0030 SHEET 8 OF 8